

SONiX 32-Bit Cortex-M0 Micro-Controller

DMA CT16B Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of CT16B in DMA mode.

2 FEATURE

- CT16B PWM function with DMA mode example.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Initialization

The CT16B0 and DMA functions are initialized in the main code. The example is to use DMA to transfer SRAM data to CT16B MR0~3. Each time TC is matched with MR9, PWM0~3 will change the duty cycle.

```

74 //-----
75 //User Code starts HERE!!!
76
77 // Init CT16B0 & DMA
78 DMACT16B0PWM0123Init();

```

The setting of DMA should refer to the table in the specifications.

```

115 void DMACT16B0PWM0123Init(void)
116 {
117     DMA_InitSt stDMACH_Init;
118
119     // Set DMA CH0 as M(SRAM) to P(CT16B0 PWM0~3) mode
120     // Source : Memory, Increment, 16bit, no request, Burst 4
121     stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_INC;
122     stDMACH_Init.b_SrcMode     = DMA_SRC_MEMORY;
123     stDMACH_Init.b_SrcWidth   = DMA_SRC_WIDTH_16BIT;
124     stDMACH_Init.b_SrcReqSel  = DMA_SRCRS_NONE;
125     stDMACH_Init.b_BurstSize  = DMA_BURST_4;
126     // Destination : Peripheral CT16B0, Increment cyclic, 16bit, CT16B0 PWM0~3
127     stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_CYC;
128     stDMACH_Init.b_DstMode     = DMA_DST_PERIPHERAL;
129     stDMACH_Init.b_DstWidth   = DMA_DST_WIDTH_16BIT;
130     stDMACH_Init.b_DstReqSel  = DMA_DSTRS_CT16B0_MR0;
131     // Channel : LV0, Enable TC/ABT interrupt
132     stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
133     stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
134     stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;
135
136     // Initial DMA
137     DMA_Init(&stDMACH_Init, eDMA_CH0);

```

The PWM period is set to 1ms, and the initial duty cycles of PWM0~3 are 10%, 30%, 70% and 90% respectively.

```

139 // Initial CT16B
140 CT16B0_Init();
141
142 //Set MR9 value for 1ms period ==> count value = 1000*12 = 12000
143 SN_CT16B0->MR9 = 12000;
144
145 //Set MR0/1/2/3 value as 10%/30%/70%/90% duty
146 SN_CT16B0->MR0 = 1200;
147 SN_CT16B0->MR1 = 3600;
148 SN_CT16B0->MR2 = 8400;
149 SN_CT16B0->MR3 = 10800;
150
151 //Enable PWM function, IOs and select the PWM modes
152 SN_CT16B0->PWMCTRL =
153     (mskCT16_PWM0EN_EN | mskCT16_PWM0MODE_2 | mskCT16_PWM0IOEN_EN) | //Enable PWM0 function, IO and select as PWM mode 1
154     (mskCT16_PWM1EN_EN | mskCT16_PWM1MODE_2 | mskCT16_PWM1IOEN_EN) | //Enable PWM1 function, IO and select as PWM mode 1
155     (mskCT16_PWM2EN_EN | mskCT16_PWM2MODE_2 | mskCT16_PWM2IOEN_EN) | //Enable PWM2 function, IO and select as PWM mode 1
156     (mskCT16_PWM3EN_EN | mskCT16_PWM3MODE_2 | mskCT16_PWM3IOEN_EN); //Enable PWM3 function, IO and select as PWM mode 1
157
158 //Set MR9 match TC reset
159 SN_CT16B0->MCTRL = mskCT16_MR9RST_EN;
160
161 //MR0 DMA request can issue.
162 SN_CT16B0->DMA = mskCT16_MR0_DMA_EN;
163
164 //Set CT16B3 as the up-counting mode.
165 SN_CT16B0->TMRCTRL = (mskCT16_CRST | mskCT16_CM_EDGE_UP);
166
167 //Wait until timer reset done.
168 while (SN_CT16B0->TMRCTRL & mskCT16_CRST);
169 }

```

4.2 DMA CT16B function enable

DMA request will be issued when the TC matches MR0, and the new MRn value will be loaded when TC=0. The DMA TC flag and interrupt will be set when all data transfers are completed.

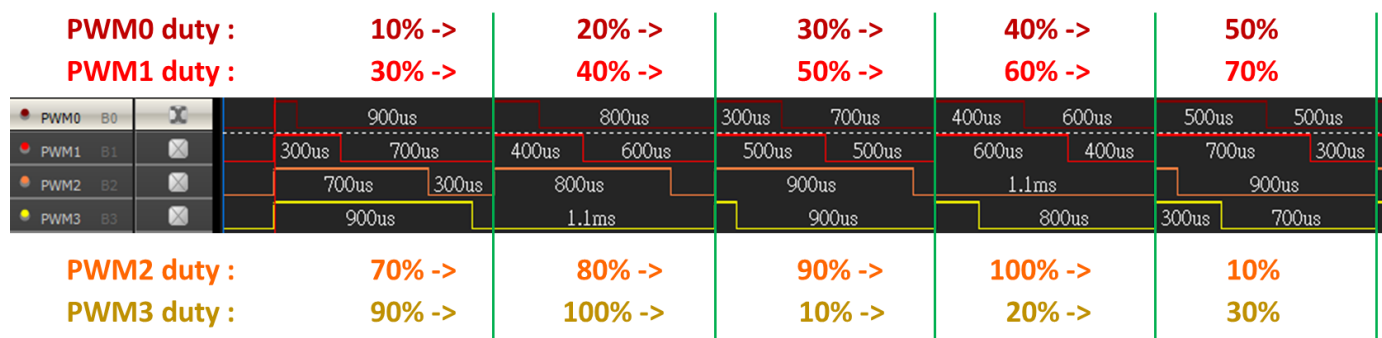
```

98 // Start DMA CT16B0 PWM0~3 transfer
99 DMACT16B0PWM0123Start();
100
101 // Wait for DMA TC interrupt flag
102 while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
103
104 while(1); // Pass
105 }

```

5 RESULT

Each time the PWM cycle is completed, the new duty value will be updated. According to the data in SRAM, the PWM duty will increase by 10% after DMA transfer. The results can be observed through PWM0~PWM3 Pins.



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